John Oyer

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EDUCATION

Johns Hopkins University, Baltimore, MD
PhD Student, Biomedical Engineering Aug 2026 – Present

University of Michigan, Ann Arbor, MI
Master of Science in Engineering, Computer Science and Engineering, GPA: 3.93/4.00 May 2026
Bachelor of Science in Engineering, Computer Science, Summa Cum Laude, GPA: 3.93/4.00 Sep 2025

Northside College Prep, Chicago, IL
High School Diploma Jun 2022

SKILLS

Technical: Python, C++, MATLAB, R; PyTorch, TensorFlow, Scikit-Learn; High-Performance Computing, Linux, Git
Areas: Computer Vision, Machine Learning, Medical Image Analysis, Clinical Data Analysis, Translational Research

RESEARCH EXPERIENCE

Michigan Medicine Department of Radiology, Ann Arbor, MI
Assistant Researcher, C. Galbán Lab Sep 2024 – May 2026
Advisors: Professor Craig Galbán, Assistant Professor Sundaresh Ram, and Professor MeiLan Han

In all projects: I designed and independently implemented machine learning algorithms, performed statistical analyses, produced figures and graphs, and evaluated methods on relevant clinical datasets.

- **AI-Based Detection of Mucus Plugs in CT Scans** *(U.S. Provisional Patent Filed)* **–** [Pub. 1, 3, 4, 5]
  - Developed a clinically and quantitatively validated machine learning algorithm for automated detection of mucus plugs in chest CT scans, addressing a key bottleneck in monitoring of COPD and asthma.
  - Addressed severe training data scarcity by leveraging airway anatomical structure as a prior, combining multiple machine learning techniques into an end-to-end detection pipeline.
  - Evaluated the method on 20,000+ CT scans across 7 cohorts spanning COPD, asthma, and healthy cases; analyses were either more extensive than prior manual studies or conducted for the first time.
  - Revealed novel associations with cause-specific mortality, lung function, quality of life, acute respiratory events, air pollution exposure, and glycoprotein overproduction, among other features; findings persisted in adjusted models and among patients without disease progression.
  - Collaborated with a multidisciplinary team of engineers, scientists, and clinicians to refine model design, interpret outcomes, and ensure clinical feasibility.
- **Machine Learning for Airway Segmentation –** [Pub. 2]
  - Enhanced deep learning airway segmentation methods via connectivity refinement, outperforming baseline and state-of-the-art approaches across multiple evaluation metrics.
  - Demonstrated improved performance in two cohorts: one of mostly healthy patients and another of patients with severe airway pathology.
- **Dictionary Learning for Lung Volume Reduction Surgery Outcome Prediction –** [Pub. 6]
  - Developed a patch-based dictionary learning model to encode CT lung volumes and predict clinical outcomes (lung function) following lung volume reduction surgery.
  - Used model classification results to successfully distinguish patients with lung function improvement from those without in a held-out test set.

PUBLICATIONS

1. **Oyer JM,** Namvar A, Labaki WW, Hoff BA, Kazerooni EA, Hatt CR, Bosma C, Martinez FJ, Han MK, Galban CJ, Ram S. Artificial Intelligence-Based Detection of Airway Mucus Plugs on CT and Associations With Clinical Outcomes in COPDGene [Preprint]. Available from: https://www.medrxiv.org/content/10.64898/2026.06.10.26355393v1
2. **Oyer JM,** Namvar A, Hoff BA, Labaki WW, Kazerooni EA, Hatt CR, Martinez FJ, Han MK, Galban CJ, Ram S. RepAir: a framework for airway segmentation and discontinuity correction in CT. arXiv:2511.14649 [Conference paper]. Oral Presentation delivered at IEEE SSIAI Conference; 2026 Mar 22–24; Santa Fe, NM. Available from: https://doi.org/10.48550/arXiv.2511.14649
3. **Oyer JM,** Namvar A, Hoff BA, Labaki WW, Kazerooni EA, Martinez FJ, Hatt CR, Han MK, Galban CJ, Ram S. AI-based detection of CT airway mucus plugs and clinical correlates in COPDGene [abstract]. Oral Presentation at ATS International Conference; 2026 May 17–20; Orlando, FL.
4. Ram S*, **Oyer JM***, Ng ASH, Labaki WW, Bosma C, Curtis JL, Kazerooni EA, Hatt CR, Paine R 3rd, Soni AV, Ritchie AI, Wedzicha JA, Bateman LA, Hoffman EA, Martinez FJ, Galban CJ, Han MK, Boucher RC, Kesimer M; SOURCE and SPIROMICS Investigators. Mucus plugging and its associations with mucin concentrations, symptoms, smoking burden, and chronic bronchitis in early COPD: SOURCE cohort analysis [abstract]. Oral Presentation delivered at ATS International Conference; 2026 May 17–20; Orlando, FL.
5. Ram S*, **Oyer JM***, Ng ASH, Labaki WW, Bosma C, Curtis JL, Kazerooni EA, Hatt CR, Paine R 3rd, Soni AV, Ritchie AI, Wedzicha JA, Bateman LA, Hoffman EA, Martinez FJ, Galban CJ, Han MK; SOURCE and SPIROMICS Investigators. Significance of mucus plugs in early COPD: an analysis of the SOURCE cohort [abstract]. Oral Presentation delivered at ATS International Conference; 2026 May 17–20; Orlando, FL.
6. Labaki WW, Lagisetty KH, Reddy RM, Meldrum CA, Wayne MT, Curtis JL, Sauler M, Bose S, Murray S, **Oyer JM**, Hatt CR, Kazerooni EA, Han MK, Galban CJ, Ram S. Chest CT-Based Machine Learning for Predicting Response to Lung Volume Reduction Surgery: A Pilot Study [abstract]. Oral presentation delivered at ERS Congress; 2026 Sep 5–9; Barcelona, Spain.

**Equal Contribution*

VOLUNTEERING

Citizens' Climate Lobby — *Treasurer & Lobby Meeting Leader* Dec 2020 – Present

- Treasurer (University of Michigan Chapter; Jan 2025 – Dec 2025): Secured $3000 through merit-based grants and fundraising, managing chapter finances to support Washington D.C. conference attendance and advocacy.
- Participated in over 5 meetings with Congressional offices, including one as meeting leader, advocating for bipartisan climate policy and contributing to measurable legislative engagement.
- Organized local events and coordinated with fellow leaders, engaging members in climate policy campaigns.

Turner Senior Resource Center — *Computer Mentor (Volunteer)* Jul 2025 – April 2026

- Mentoring seniors weekly in basic computer and software skills, fostering digital literacy.

HONORS & AWARDS

Early Career Professional Highlight — *ATS RSF Assembly* Mar 2026

- Selected as monthly ECP feature of American Thoracic Society Respiratory Structure & Function Assembly.

Empowering Research with AI Award — *AI Institutes at Michigan* Mar 2026

- Merit-based $1000 award recognizing AI-based mucus plug detection research.

Abstract Scholarship — *ATS* Mar 2026

- Merit-based award covering full cost of 2026 American Thoracic Society International Conference registration.

Rackham Conference Travel Grant — *University of Michigan* Jan 2026

- $900 award for 2026 SSIAI conference attendance.

Tau Beta Pi (Engineering Honor Society) — *Elected to University of Michigan Chapter* Dec 2023